Year 13 Further Maths - Applications of calculus

Similar questions - Proof by induction

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A function is defined by $f(t) = \ln(1+at)$ where a is a constant.

(a) By considering $f'(t)$, $f''(t)$, $f^{(3)}(t)$ and $f^{(4)}(t)$, make a conjecture for $f^{(n)}(t)$ in terms of n and a for integers $n \geqslant 1$. **[3]**

(b) Prove your conjecture by induction. **[4]**

(c) When $a = 2$, find the rate at which the 6th derivative of f is changing when $t = \frac{3}{2}$. **[2]**

2 Prove by induction that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for all integers $n \geqslant 0$. **[5]**

Answers

Question 1

(a) $f' = \dfrac{a}{1+at}$, $f'' = -\dfrac{a^2}{(1+at)^2}$, $f^{(3)} = \dfrac{2a^3}{(1+at)^3}$, $f^{(4)} = -\dfrac{6a^4}{(1+at)^4}$. Conjecture: $f^{(n)}(t) = \dfrac{(-1)^{n-1}(n-1)!\,a^n}{(1+at)^n}$

(b) $n=1$: $\dfrac{(-1)^0 0!\,a}{1+at} = \dfrac{a}{1+at}$, true. Assume true for $n=k$: $f^{(k)} = (-1)^{k-1}(k-1)!\,a^k(1+at)^{-k}$. Then $f^{(k+1)} = (-1)^{k-1}(k-1)!\,a^k \cdot (-k)\,a\,(1+at)^{-k-1} = \dfrac{(-1)^k\,k!\,a^{k+1}}{(1+at)^{k+1}}$, the formula for $n=k+1$. True for $n=1$ and $k \Rightarrow k+1$, so true for all $n \geqslant 1$

(c) The rate of change of the 6th derivative is the 7th derivative: $f^{(7)} = \dfrac{6!\,2^7}{(1+2t)^7}$. At $t=\frac{3}{2}$: $\dfrac{720\times 128}{4^7} = \dfrac{92160}{16384} = \dfrac{45}{8} = 5.625$

Question 2

$n=0$: $3+4=7$, divisible by 7. Assume $3^{2k+1}+2^{k+2} = 7m$. Then $3^{2k+3}+2^{k+3} = 9\times 3^{2k+1} + 2\times 2^{k+2} = 9\left(7m-2^{k+2}\right) + 2\times 2^{k+2} = 63m - 7\times 2^{k+2} = 7\left(9m-2^{k+2}\right)$, divisible by 7. True for $n=0$ and $k \Rightarrow k+1$, so true for all $n \geqslant 0$